COMPUTER NETWORKS: COMPLETE EXAM CRAM GUIDE

Comprehensive 20-Minute Master Review | Lectures 01 through 10 | Definitions, Formulas & Key Tables

LECTURE 01: INTRODUCTION TO DATA COMMUNICATION

1. Key Definitions (One-Liners):

- **Data Communication:** The exchange of data (in form of 0s and 1s) between two devices via a physical or wireless transmission medium.
- **Telecommunication:** Communication over a distance including telephony, telegraphy, and television.
- **Data Communication System:** An integrated combination of hardware (physical equipment) and software (programs) designed to transmit information.

2. Five Essential Components of Data Communication:

Component	One-Liner Definition / Role	Examples
Message	The information or data to be communicated.	Text, numbers, pictures, audio, video.
Sender (Source)	The device that generates and transmits the message.	Computer, workstation, telephone handset, camera.
Receiver (Destination)	The device that receives and processes the message.	Computer, printer, television set.
Medium	The physical path over which the message travels.	Twisted-pair wire, coaxial cable, fiber optic, radio waves.
Protocol	A set of rules that governs data communication.	HTTP, TCP/IP, Ethernet rules.

3. Fundamental Data Representation Formats:

- **Text:** Represented as bit patterns using standards like ASCII (7-bit), Extended ASCII (8-bit), or Unicode (16/32-bit).
- **Numbers:** Converted directly to positional binary numbers (base-2) for mathematical storage.
- **Images:** Matrix of pixels; each pixel color depth determines required bits (e.g., 24-bit TrueColor).
- **Audio:** Continuous sound waves converted into discrete digital signals via sampling and quantization.
- **Video:** Recording/broadcasting of motion, formed as continuous entities or sequences of discrete frame images.

4. Four Fundamental Criteria for Effective Data Communication:

1. **Delivery:** System must deliver data to the correct destination accurately.
2. **Accuracy:** System must deliver data unaltered (corrupted data is unusable).
3. **Timeliness:** System must deliver data in real-time without excessive delay.
4. **Jitter:** Uneven variation in packet arrival time (crucial for real-time audio/video).

LECTURE 02: NETWORK CONCEPTS, TERMINOLOGY & DATA FLOW

1. Essential Network Terminology:

Term	Exact Exam Definition
Network	An interconnected collection of autonomous computers and communication devices sharing resources.
Node	Any device attached to a network capable of sending or receiving data (e.g., PC, printer, router).

Term	Exact Exam Definition
Link	The physical or wireless communication pathway connecting two adjacent nodes.
Path	The complete end-to-end sequence of links traversed to reach a final destination.
Circuit	A dedicated physical path or channel configured over which data travels.
Session	An active, ongoing communication dialogue between two network entities/applications.
Packetizing	The process of dividing a long message into smaller, fixed-length packets prior to transmission.
Routing	The algorithmic determination of the optimal path for packets from source to destination.

2. Directional Transmission Modes (Data Flow):

• **Simplex Mode:** Strictly unidirectional data flow (one device only transmits, the other only receives). Entire link capacity is dedicated to one direction. *Examples:* Keyboard to CPU, TV broadcast, traditional monitor.

• **Half-Duplex Mode:** Bidirectional data flow, but only ONE direction at a time. Each station can transmit and receive, but not simultaneously. Entire link capacity is shared temporally. *Examples:* Walkie-talkies, CB radio.

• **Full-Duplex Mode:** Simultaneous bidirectional data flow. Link signals share capacity either by physical separation or frequency splitting. *Examples:* Telephone conversation, modern switched Ethernet.

3. Distributed Processing & Three Core Network Criteria:

• **Distributed Processing:** Dividing a massive computational process among multiple individual computers across a network rather than relying on one central machine.

> **EXAM FOCUS — THREE NETWORK CRITERIA:**
> 1. **Performance:** Measured by *Transit Time* (time required for message to travel) and *Response Time* (elapsed time between inquiry & response). Depends on: Number of users, Transmission medium, Hardware capabilities, Software efficiency.
> 2. **Reliability:** Measured by Frequency of failure, Recovery time after failure, and Robustness in catastrophes (fire, earthquake, theft).
> 3. **Security:** Protecting data from unauthorized access/alteration, preventing damage, and maintaining disaster recovery procedures.

LECTURE 03: LINE CONFIGURATIONS & PHYSICAL TOPOLOGIES

1. Line Configuration Types:

• **Point-to-Point:** Dedicated link between exactly two devices. Entire link capacity is reserved for those two devices (e.g., infrared remote to TV, microwave link).

• **Multipoint (Multidrop):** More than two specific devices share a single link. Capacity is shared spatially (simultaneous) or temporally (time-shared).

2. Master Comparison of Physical Topologies:

Topology	Structure / Formula	Key Advantages	Key Disadvantages / Exam Traps
Bus	Single main central backbone cable with drop lines & T-connectors.	Easy to install, uses minimal cabling.	Backbone failure kills entire network. Requires **BNC Barrel Terminators** at open ends.
Mesh (True)	Fully connected point-to-point links. **Links = n(n-1)/2** **I/O Ports = n-1**	Dedicated links eliminate traffic; highly robust, secure, easy fault isolation.	Prohibitively expensive cabling & I/O ports; sheer bulk of wiring is difficult to install.
Star	Devices connect to central controller (Hub/Switch). **Links = n**	Less expensive than mesh; easy setup; one link failure affects only that node.	**Single point of failure: If central hub fails, whole network dies. More cable than bus.**

Topology	Structure / Formula	Key Advantages	Key Disadvantages / Exam Traps
Ring	Unidirectional loop; each node connects to 2 neighbors via repeaters.	Easy to install/reconfigure; traffic degrades gracefully under heavy load.	Unidirectional break kills whole network. Adding/removing nodes disrupts traffic.
Tree	Hierarchical star topology connected to central hub/backbone.	Scalable; easy segment expansion & isolation.	Hub dependency; extensive cabling.
Hybrid	Combination of 2+ topologies (e.g., Star backbone with 3 Bus LANs).	Flexible; leverages strengths of different topologies.	Complex design and maintenance.

NUMERICAL FORMULAS (MUST MEMORIZE):
• For a Fully Connected Mesh with **n** nodes:
— Total Physical Links required = **n(n - 1) / 2**
— Number of I/O Ports required per device = **n - 1**
Example: 6 devices require **6(5)/2 = 15 links** and **5 I/O ports** per device.
• **BNC Barrel Connector:** Used in Bus topologies to join two cable segments to extend length or repair a break; absorbs energy.

LECTURE 04: CATEGORIES OF NETWORKS & INTERNETWORKS

1. Network Classification by Geographic Scale:

Category	Full Name	Geographic Span / Coverage	Typical Ownership & Features
PAN	Personal Area Network	Under 10 meters (within a room)	Personal devices (Bluetooth headphones, smartwatch, phone sync).
LAN	Local Area Network	Single office, building, or campus	Privately owned, high data rate, low error rate (Ethernet, Wi-Fi).
MAN	Metropolitan Area Network	Entire city or town	Cable TV network, city-wide corporate/municipal network.
WAN	Wide Area Network	Country, continent, or globe	Public/telecom providers, connects LANs across huge distances.

2. Internetworks & The Internet:

• **Internetwork (internet):** Formed when two or more distinct networks (LANs, WANs) are connected via routers/gateways to exchange data.

• **The Internet:** The global, publicly accessible internetwork connecting millions of private, public, academic, and business networks worldwide.

LECTURE 05: THE OSI REFERENCE MODEL (7 LAYERS)

1. OSI Model Overview:

The **Open Systems Interconnection (OSI)** model is a 7-layer theoretical framework established by ISO for standardized protocol design.

L#	Layer Name	Data Unit	Primary Functions & Exam Buzzwords
7	Application	Data	Network services to end-user applications (HTTP, FTP, SMTP, DNS). Direct user interface.
6	Presentation	Data	**Syntax & Semantics: Translation (code conversion), Encryption/Decryption, Data Compression.**

L#	Layer Name	Data Unit	Primary Functions & Exam Buzzwords
5	Session	Data	**Dialog Control: Session creation/maintenance/termination, Synchronization (checkpoints).**
4	Transport	Segment	**End-to-End Delivery: Service-point (Port) addressing, Segmentation & Reassembly, Connection control, Flow & Error control.**
3	Network	Packet	**Source-to-Destination Delivery: Logical (IP) Addressing, Routing across heterogeneous networks.**
2	Data Link	Frame	**Hop-to-Hop (Node-to-Node) Delivery: Physical (MAC) Addressing, Framing, Flow control, Error control, Media Access Control.**
1	Physical	Bits	**Bit-by-Bit Transmission: Mechanical/electrical specs, Bit rate control, Synchronization, Line configuration.**

EXAM MNEMONIC & KEY DISTINCTION:
• Mnemonic (Bottom-to-Top): **P**lease **D**o **N**ot **T**ouch **S**teve's **P**et **A**lligator (Physical, Data Link, Network, Transport, Session, Presentation, Application).
• **Data Link vs. Transport Layer:** Data Link ensures *hop-to-hop* delivery between adjacent nodes; Transport ensures *end-to-end (process-to-process)* delivery from ultimate source to ultimate destination.

LECTURE 06: TCP/IP PROTOCOL SUITE & ADDRESSING

1. TCP/IP vs. OSI Model Mapping:

Developed by DoD (DARPA) and maintained by IETF. TCP/IP consolidates OSI's top 3 layers (5, 6, 7) into a single **Application Layer**, and lower 2 layers into **Network Access**.

TCP/IP Layer	Corresponding OSI Layers	Core Protocols In Suite
Application	Application, Presentation, Session	HTTP, HTTPS, FTP, SMTP, DNS, SNMP, TELNET, MIME.
Transport	Transport	**TCP (Connection-oriented, reliable) & UDP (Connectionless, fast), SCTP.**
Internet (Network)	Network	**IP (IPv4/IPv6), ICMP, IGMP, ARP, RARP.**
Network Access	Data Link, Physical	Ethernet, Wi-Fi (802.11), Point-to-Point (PPP), Token Ring.

2. Four Levels of Addressing in TCP/IP:

• **Physical (MAC) Address:** 48-bit (6-byte) hardware address burned into NIC. Used for node-to-node hop delivery at Layer 2 (e.g., 00:1A:2B:3C:4D:5E).
• **Logical (IP) Address:** Universal 32-bit (IPv4) or 128-bit (IPv6) network address used for global end-to-end routing at Layer 3.
• **Port Address:** 16-bit integer (0 to 65535) identifying specific application processes on a host at Layer 4 (e.g., HTTP = Port 80, HTTPS = 443, FTP = 21).
• **Specific Address:** User-friendly application-specific aliases (e.g., URL www.google.com, email address).

LECTURE 07: GUIDED TRANSMISSION MEDIA (WIRED)

1. Guided Media Characteristics & Comparison:

Cable Type	Physical Structure	Key Sub-types / Standards	Advantages & Disadvantages
Twisted-Pair	Two insulated copper conductors twisted together to reduce noise & crosstalk.	**UTP: Unshielded (cheap, common Cat5e/Cat6).** **STP: Shielded (metal foil casing, better EMI protection).**	**Pros: Inexpensive, flexible, easy installation.** **Cons: High attenuation, susceptible to EMI.**
Coaxial Cable	Central copper core, insulating layer, metallic braid shield, outer jacket.	50-ohm (digital baseband) & 75-ohm (analog broadband). Connectors: BNC.	**Pros: Higher bandwidth & EMI shield than UTP.** **Cons: Bulkier, higher cost than UTP.**
Fiber-Optic	Glass/plastic core surrounded by lower-density glass cladding (Total Internal Reflection).	**Multimode Step-Index, Multimode Graded-Index, Single-Mode (small core, laser).**	**Pros: Massive bandwidth, zero EMI, low attenuation, high security.** **Cons: Expensive, fragile, complex splicing.**

LECTURE 08: UNGUIDED TRANSMISSION MEDIA (WIRELESS)

1. Wireless Signal Propagation Methods:

- **Ground Propagation (< 2 MHz):** Low-frequency signals hug the Earth's curvature in all directions through lower atmosphere.
- **Sky Propagation (2 MHz – 30 MHz):** High-frequency signals radiate into the **ionosphere** and bounce back down to Earth for global long-distance reach.
- **Line-of-Sight (LOS) Propagation (> 30 MHz):** Very high-frequency signals travel in direct straight lines between aligned, unobstructed antennas.

2. Wireless Bands & Microwave Characteristics:

Band Category	Frequency Range	Directionality & Trait	Typical Applications
Radio Waves	3 kHz – 1 GHz	**Omnidirectional: Spreads in all directions, penetrates walls.**	AM/FM radio, TV broadcast, cordless phones, paging.
Microwaves	1 GHz – 300 GHz	**Unidirectional: Narrow focused beam, Line-of-Sight, cannot penetrate walls.**	Cellular networks, satellite links, wireless LANs.
Infrared	300 GHz – 400 GHz	**Short-range LOS: Blocked by walls (no room interference), sunlight sensitive.**	TV remotes, wireless mice/keyboards, IrDA ports.

MICROWAVE ANTENNAS & SATELLITES:

- **Parabolic Dish Antenna:** Uses parabolic reflection where parallel incoming waves reflect and intersect at a single point called the **Focus**.
- **Horn Antenna:** Looks like a giant scoop; deflects outgoing signals into parallel beams and scoops incoming signals down into the stem.
- **Satellite Communications:** Orbiting satellite at ~22,000 miles (36,000 km) acts as a high-altitude repeater linking earth stations.

LECTURES 09 & 10: MULTIPLEXING TECHNIQUES

1. Concept & Hardware:

Multiplexing allows multiple signals to share a single high-capacity data link simultaneously. Executed by a **Multiplexer (MUX)** at sending end and **Demultiplexer (DEMUX)** at receiving end.

2. Comparison of The 3 Major Multiplexing Categories:

Technique	Signal Type	Mechanism / Core Principle	Key Features / Exam Traps
FDM	Analog	Splits total link bandwidth into distinct frequency channels.	Requires **Guard Bands** (unused frequency strips) to prevent crosstalk.

Technique	Signal Type	Mechanism / Core Principle	Key Features / Exam Traps
WDM	Analog (Optical)	Combines different wavelengths (colors) of light over fiber.	Uses optical **Prisms** to combine at MUX and refract/split at DEMUX.
Synchronous TDM	Digital	Divides time into fixed slots; devices take turns in round-robin order.	Wastes bandwidth with **Empty Slots** if a device is idle. No addresses needed.
Statistical TDM	Digital	Dynamically allocates slots only to active devices with data.	Eliminates empty slots. Requires **Destination Address** inside every slot.

3. Synchronous TDM In-Depth Mechanics:

• **Interleaving:** The MUX takes 1 data unit (bit, character, block) from each input sequentially to construct a complete **Frame**.

• **Time Slot Duration Math:** If input unit time slot is **T sec** and there are **n** connections, the link slot duration is **T / n** (data travels **n times faster** on link).

• **Data Rate Management (Handling Mismatched Input Speeds):**

1. *Multilevel Multiplexing:* Combines lower-speed lines first using a sub-MUX until their speed matches higher-speed lines.
2. *Multiple-Slot Allocation:* Allocates multiple time slots in each frame to a faster input line.
3. *Pulse Stuffing (Bit Padding):* Inserts dummy bits into slower lines to artificially raise their speed to match the dominant rate.

4. Statistical TDM Addressing Formula:

Because slots are assigned dynamically, each slot must carry data + destination address. For **N** output lines, the address requires **n = $\log_2$ N** bits.

> **20-MINUTE PRE-EXAM QUICK-CHECK SUMMARY:**
> ✓ **Mesh Links Formula:** n(n-1)/2 | **Mesh Ports:** n-1
> ✓ **5 Components:** Message, Sender, Receiver, Medium, Protocol
> ✓ **OSI Layers (1-7):** Physical, Data Link, Network, Transport, Session, Presentation, Application
> ✓ **Addressing:** MAC (Layer 2, 48-bit), IP (Layer 3, 32/128-bit), Port (Layer 4, 16-bit)
> ✓ **Multiplexing:** FDM = Frequency + Guard Bands | WDM = Light + Prisms | TDM = Time Slots (Sync = Fixed/Empty Slots; Stat = Dynamic + Address)